1.07 Algebra: Simultaneous Equations

Chapter 3 · Original slide 3 · Solutions sets

The solution(s) to an equation may be:

$2x + 1 = 5$

$x^2 + 3x + 2 = 0$

$x > 3$

$x^2 = -1$

$x^2 + x = x\,(x + 1)$

! The solutions to an equation are known as the solution set.

A single value:

Multiple values:

An infinitely large set of values:

No (real) values!

Every value!

The point is that you shouldn't think of the solution to an equation/inequality as an 'answer', but a set of values, which might just be a set of 1 value (known as a singleton set), a set of no values (i.e. the empty set $\varnothing$), or an infinite set (in the last example above, this was $\mathbb{R}$)

Chapter 3 · Original slide 4 · Solutions sets

Scenario	Example	Solution Set
	$x + y = 9$ $x - y = 1$	
No solutions:	$x + y = 1$ $x + y = 3$	
	$x + y = 1$ $2x + 2y = 2$	

For simultaneous equations, the same is true, except each 'solution' in the solution set is an assignment to multiple variables.

All equations have to be satisfied at the same time, i.e. 'simultaneously'.

A single solution:

$x^2 + y^2 = 10$
$x + y = 4$

Two solutions:

Infinitely large set of solutions:

1.07 Algebra: Simultaneous Equations

Chapter 3 · Original slide 5 · 1 :: Simultaneous Equations

Recap!

Solve the simultaneous equations:

$3x + y = 8$
$2x - 3y = 9$

Solve the simultaneous equations:

$x + 2y = 3$
$x^2 + 3xy = 10$

We can’t use elimination this time as nothing would cancel.

We instead:

(1) Rearrange linear equation to make x or y the subject.

(2) Substitute into quadratic equation and solve.

We can either use substitution (i.e. making x or y the subject of one equation, and substituting it into the other) or elimination, but the latter is easier for linear equations.

Chapter 3 · Original slide 6 · Test Your Understanding

Solve the simultaneous equations:

$3x^2 + y^2 = 21$
$y = x + 1$

1.07 Algebra: Simultaneous Equations

Chapter 3 · Original slide 7 · Exercise 3A/B – Y1 Pages 40, 41

4 Solve the simultaneous equations:
$x + y = 3, \quad x^2 - 3y = 1$

Chapter 3 · Original slide 8 · Exercise 3A/B – Y1 Pages 40, 41

5 a By eliminating y from the equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, show that $x^2 - 2x - 11 = 0$.
b Hence, or otherwise, solve the simultaneous equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, giving your answers in the form $a \pm b\sqrt{3}$, where a,b are integers.

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b Find the second pair of solutions for the simultaneous equations.

Chapter 3 · Original slide 10 · Extension

1

[MAT 2012 1G] There are *positive* real numbers x and y which solve the equations

$2x + ky = 4,$
$x + y = k$

for:

A) All values of k;

B) No values of k;

C) $k = 2$ only;

D) Only $k > -2$

2

[STEP 2010 Q1] Given that

$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$

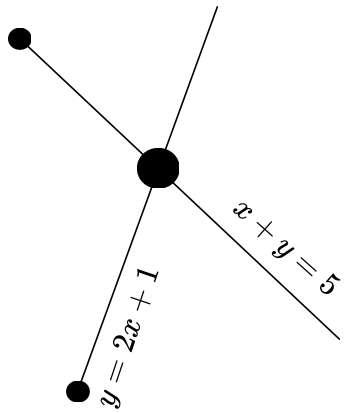
a) Find the values of a, b, c, d.

b) Solve the simultaneous equations:

$5x^2 + 2y^2 - 6xy + 4x - 4y = 9,$
$6x^2 + 3y^2 - 8xy + 8x - 8y = 14$
(Hint: Can we use the same method in (a) to rewrite the second equation?)

1.07 Algebra: Simultaneous Equations

Chapter 3 · Original slide 11 · Simultaneous Equations and Graphs



Now suppose we introduced a second simultaneous equation:

$y = 2x + 1$

$x + y = 5$

The second line again consists of all points (x,y) which satisfy the equation. So what point must satisfy both equations simultaneously?

Recall that a line with a given equation is the set of all points that satisfy the equation.

i.e. It is a graphical representation of the solution set where each (x, y) point represents each of the solutions x and y to the equation. e.g. $x = 3, y = 7$

Chapter 3 · Original slide 12 · Example

a) On the same axes, draw the graphs of:

$2x + y = 3$
$y = x^2 - 3x + 1$

b) Use your graph to write down the solutions to the simultaneous equations.

(We could always substitute into the original equations to check they work)

c) What algebraic method (perhaps thinking about the previous chapter), could we have used to show the graphs would have intersected twice?

1.07 Algebra: Simultaneous Equations

Chapter 3 · Original slide 13 · Another Example

a) On the same axes, draw the graphs of:

$y = 2x - 2$
$y = x^2 + 4x + 1$

b) Prove algebraically that the lines never meet.

Chapter 3 · Original slide 14 · Final Example

a b

The line with equation $y = 2x + 1$ meets the curve with equation

$kx^2 + 2y + (k - 2) = 0$ at exactly one point. Given that k is a positive constant:

a) Find the value of k.

b) For this value of k, find the coordinates of this point of intersection.

1.07 Algebra: Simultaneous Equations

Chapter 3 · Original slide 15 · Exercise 3C – Y1 Page 45

8 Given the simultaneous equations $2x - y = 1, \quad x^2 + 4ky + 5k = 0$, where k is a non-zero constant,

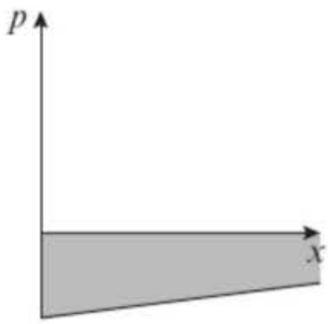
a show that $x^2 + 8kx + k = 0$.

Given that $x^2 + 8kx + k = 0$ has equal roots,

b find the value of k

c for this value of k, find the solution of the simultaneous equations.

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(5 marks)

9 A swimmer dives into a pool. Her position, p m, underwater can be modelled in relation to her horizontal distance, x m, from the point she entered the water as a quadratic equation $p = \frac{1}{2}x^2 - 3x$. The position of the bottom of the pool can be modelled by the linear equation $p = 0.3x - 6$. Determine whether this model predicts that the swimmer will touch the bottom of the pool.